Technical report on the second literal assertion in Lee's Lemma 19

Haiman counterexample audit project

2026-06-28

Abstract

This report expands the short section "A second literal assertion fails" in the project documentation. The purpose is narrow: explain exactly what was computed, which representation-theoretic facts are being used, why no Borel choice is needed for this particular negative test, and what remains open.

The current audit proves that one printed sentence in Lee's Lemma 19 is false if read literally: not every 89×89 minor belongs to one of the 15 degree-89 predecessor modules. It also now records the more precise minimality-relevant list: the 15 Schur modules $V = S_\lambda W$ for which $V \otimes S_\mu W$ can contribute to the target $S_\nu W$, at the Littlewood–Richardson/tensor-product level. This is a serious gap in the written proof, but it is not by itself a proof that the claimed degree-90 generator is nonminimal, nor a proof that Lee's theorem is false.

1 Executive summary

The source paper claims that a degree-90 Schur module

$$S_\nu W, \qquad \nu = (133, 130, 126, 122, 119, 60, 60, 60),$$

occurs among the minimal generators of J_8, the ideal of the principal component $P_8 \subset \mathrm{Hilb}^9(\mathbb{C}^8)$ near $\mathfrak{m}^2$. Lemma 19 is the computational step intended to prove this.

The audit has reconstructed Lee's 90×115 matrix M and certified a nonzero 90×90 minor. The section studied here concerns a different sentence on the degree-89 side. The paper says, in effect, that each 89×89 minor belongs to one of 15 modules obtained by a Littlewood–Richardson search. The audit produced an explicit counterexample to that literal sentence.

There are two separate statements here, and they must not be conflated:

1. the 15 Littlewood–Richardson/tensor candidates are the only Schur-highest weights λ presently certified for a degree-89 module $V = S_\lambda W$ whose product with $S_\mu W$ can contain $S_\nu W$;

2. the displayed 89×89 minor is outside those 15 modules, so it refutes the paper's broad wording, but it is not a candidate V that can prove nonminimality of the target $S_\nu W$.

Evidence level	Statement
Computation	Deleting row 0 and the first pivot column from the certified full-rank 90×90 pivot minor gives a nonzero 89×89 determinant: $$421057 \pmod{1000003}.$$
Computation	The same minor is torus-weight homogeneous of weight $$(60, 59, 59, 140, 123, 126, 119, 115),$$ whose dominant reordering is $$(140, 126, 123, 119, 115, 60, 59, 59).$$
Computation	The 15 certified degree-89 predecessor partitions all have first part at most 132.
Theorem	Every weight of the irreducible $\mathrm{GL}(W)$-module $S_\lambda W$ has dominant representative dominated by λ. In particular, its first part is at most λ_1.
Conclusion	The explicit nonzero 89×89 minor is not a weight vector in any of the 15 listed degree-89 modules. Hence the literal sentence "each 89×89 minor belongs to one of the 15 modules" is false as written.

2 Why torus weights are enough here

The user's concern is exactly the right one: the published webpage section is short and mentions torus weights, while highest-weight statements usually involve a Borel subgroup. The distinction is as follows.

2.1 Positive membership needs more than a weight

To prove that a vector generates or lies in a particular copy of $S_\lambda W$, a torus weight alone is not sufficient. One normally also checks a highest-weight condition for a chosen Borel, or computes the projection to the desired isotypic component. This is why the audit does not claim that the weight-compatible 90×90 minor proves Lee's Schur-module assertion.

Indeed, the audit found a nonzero 90×90 minor of the claimed torus weight

$$(60, 60, 60, 133, 130, 126, 122, 119),$$

with determinant $300834 \pmod{1000003}$. But for the Borel order

$$4 < 5 < 6 < 7 < 8 < 1 < 2 < 3,$$

five simple raising derivatives are nonzero:

operator	E_{45}	E_{56}	E_{67}	E_{78}	E_{23}
residue mod 1000003	685026	176188	140239	78485	605583.

Thus that selected determinant is not itself a highest-weight vector. This is a negative computation about one determinant only; it does not rule out a different linear combination or a nonzero projection.

2.2 Negative membership can be disproved by a missing weight

The 89×89 statement is different. We are not trying to prove positive membership in one of the 15 modules. We are disproving membership by showing that the vector has a torus weight that cannot occur in any of them.

Let $T \subset \mathrm{GL}(W)$ be the diagonal torus. Any finite-dimensional rational $\mathrm{GL}(W)$-module decomposes into T-weight spaces. An irreducible $S_\lambda W$ has only weights whose dominant representative is dominated by λ. This is a standard highest-weight theorem, but the test itself does not require choosing a Borel after the list of highest weights is fixed.

Therefore, if a homogeneous determinant f has torus weight α, and the dominant reordering α^+ is not dominated by any of the 15 candidate highest weights λ, then f cannot be an element of their direct sum. It cannot be hidden there under another Borel convention; the torus weight is already impossible.

3 The reconstructed matrix and the certified minor

Lee's construction isolates the 45 variables

$$A = \{p_{r,st} : 1 \le r \le 3,\ 4 \le s \le t \le 8\}.$$

After centering the complementary variables, the selected equations have the form

$$C = M(A)z,$$

where M is a 90×115 matrix with entries $0, \pm p_{r,st}$.

The matrix reconstruction is not a transcription of entries. It is generated from the printed index formulas. The canonical sparse manifest has:

$$\begin{aligned} \text{shape} &= 90 \times 115, \\ \text{coefficient variables} &= 45, \\ \text{nonzero entries} &= 1410. \end{aligned}$$

The payload SHA-256 is

`dc4bb49eaab862c6f11ce83b9129fd9c16adc6b6d4e591fc4be8ea972d9176ac.`

At the recorded specialization modulo $p = 1000003$, the matrix has rank 90. The pivot minor has determinant

$$970351 \pmod{1000003}.$$

This proves that the corresponding integer determinant polynomial is not zero over characteristic zero.

For the degree-89 test, the audit deletes:

$$\text{row } 0, \qquad \text{pivot-column position } 0 \quad (\text{global column } 0)$$

from the same certified pivot minor. The resulting 89×89 determinant is

$$421057 \pmod{1000003}.$$

Thus this 89×89 determinant is also a nonzero polynomial in characteristic zero.

4 Weight calculation

The coordinate $p_{r,st}$ has torus weight

$$\mathrm{wt}(p_{r,st}) = (1, \ldots, 1) - e_r + e_s + e_t.$$

The matrix entries are compatible with row and column weights: an entry in row ρ, column γ, and coefficient variable a satisfies

$$\mathrm{wt}(a) = \mathrm{wt}(\rho) - \mathrm{wt}(\gamma).$$

Consequently every nonzero monomial in a fixed minor determinant has the same total torus weight:

$$\mathrm{wt}(\det M_{R,C}) = \sum_{\rho \in R} \mathrm{wt}(\rho) - \sum_{\gamma \in C} \mathrm{wt}(\gamma).$$

For the certified 89×89 minor this gives

$$\alpha = (60, 59, 59, 140, 123, 126, 119, 115).$$

Reordering into the dominant chamber gives

$$\alpha^+ = (140, 126, 123, 119, 115, 60, 59, 59).$$

5 The 15 predecessor partitions

The audit reconstructs the Littlewood–Richardson search appearing in Lee's Lemma 19. Let

$$\mu = (3, 1, 1, 1, 1, 1, 1, 0), \qquad \nu = (133, 130, 126, 122, 119, 60, 60, 60).$$

The certificate records:

- 102 partitions λ with $c^{\nu}_{\lambda,\mu} > 0$;
- 87 excluded by tensor-product dominance for the 89th tensor power;
- 15 remaining candidates, each with an explicit 88-step nonzero Littlewood–Richardson chain proving tensor occurrence.

The 15 endpoint partitions are:

#	λ
1	$(130, 129, 125, 121, 118, 60, 59, 59)$
2	$(131, 128, 125, 121, 118, 60, 59, 59)$
3	$(131, 129, 124, 121, 118, 60, 59, 59)$
4	$(131, 129, 125, 120, 118, 60, 59, 59)$
5	$(131, 129, 125, 121, 117, 60, 59, 59)$
6	$(132, 127, 125, 121, 118, 60, 59, 59)$
7	$(132, 128, 124, 121, 118, 60, 59, 59)$
8	$(132, 128, 125, 120, 118, 60, 59, 59)$
9	$(132, 128, 125, 121, 117, 60, 59, 59)$

10	$(132, 129, 123, 121, 118, 60, 59, 59)$
11	$(132, 129, 124, 120, 118, 60, 59, 59)$
12	$(132, 129, 124, 121, 117, 60, 59, 59)$
13	$(132, 129, 125, 119, 118, 60, 59, 59)$
14	$(132, 129, 125, 120, 117, 60, 59, 59)$
15	$(132, 129, 125, 121, 116, 60, 59, 59)$

Every one has $\lambda_1 \leq 132$.

5.1 Answer to the requested V-production step

At the level currently certified, the possible degree-89 Schur modules are exactly

$$V_\lambda = S_\lambda W$$

for the 15 λ's in the table above. For each one the audit checks

$$c^{\nu}_{\lambda,\mu} = 1$$

and supplies an explicit chain of 88 nonzero Littlewood–Richardson coefficients proving occurrence in the tensor power

$$S_\lambda W \subset (S_\mu W)^{\otimes 89}.$$

Consequently each listed V_λ is a necessary tensor-level candidate for a degree-89 source whose multiplication by degree-one variables can contribute to $S_\nu W$ in degree 90.

This is precisely what the file

`results/certificates/lemma19_minimality_relevant_modules.json`

records. It deliberately labels two further tasks as open:

1. construct the actual copies of these modules inside $\mathrm{Sym}^{89}(S_\mu W)$, not merely inside the tensor power;
2. test the corresponding embedded copies against J_8, or compute their multiplication image in the $S_\nu W$-isotypic component of degree 90.

Thus the audit can now produce the candidate highest weights for V, but it has not yet produced the embedded submodules whose J_8-membership would settle Lee's minimality assertion.

5.2 Current status of the symmetric-power multiplicity computation

The next representation-theoretic gate is the multiplicity

$$[S_\lambda W : \mathrm{Sym}^{89}(S_\mu W)]$$

for the 15 candidates above. The current repository reduces these numbers to explicit GL5 plethysm coefficients recorded in

`results/certificates/lemma19_symmetric_multiplicity_targets.json`.

The accepted certificate is now

`results/certificates/lemma19_symmetric_multiplicities.json`.

It proves nonzero residues modulo 1000000007 for all 15 degree-89 candidates and also for the degree-90 target.

The Sage route

`artifacts/sage/compute_symmetric_multiplicities.sage`

uses Jacobi–Trudi and the symmetric-matrix Cauchy identity

$$h_k(\mathrm{Sym}^2 V) = \mathrm{Sym}^k(\mathrm{Sym}^2 V) = \bigoplus_{\lambda \vdash k,\ \ell(\lambda) \leq 5} S_{2\lambda} V$$

to turn the GL5 plethysm coefficients into ordinary Littlewood–Richardson products. This is a theorem-level reduction, but it is currently only a secondary implementation route: the direct summation did not return the first target within 90 seconds on the local machine.

The Rust route

`artifacts/rust/src/bin/plethysm_multiplicity.rs`

uses a modular semistandard-column dynamic program and then extracts Schur multiplicities with the Weyl character formula

$$m_\alpha = \sum_{\sigma \in S_5} \mathrm{sgn}(\sigma)\, M(\alpha + \rho - \sigma\rho), \qquad \rho = (4, 3, 2, 1, 0).$$

After changing residue storage from 64-bit to 32-bit arithmetic, the dense weight-DP completed locally in about 389 seconds and wrote the certificate above. Since every recorded residue is nonzero modulo 1000000007, every corresponding integer multiplicity is nonzero.

Therefore the symmetric-power occurrence question is closed at the nonvanishing level. The remaining tasks are to construct the relevant highest-weight vectors and test their J_8-membership.

6 The actual contradiction to the literal sentence

For a weight α to occur in $S_\lambda W$, its dominant reordering α^+ must be dominated by λ. Dominance begins with the first prefix inequality:

$$\alpha_1^+ \leq \lambda_1.$$

For the certified 89×89 minor:

$$\alpha_1^+ = 140.$$

For every one of the 15 predecessor candidates:

$$\lambda_1 \leq 132.$$

Therefore

$$\alpha^+ \not\leq \lambda$$

for all 15 candidates. The determinant cannot be a weight vector in any of the 15 modules. Since the determinant is nonzero, this is a concrete counterexample to the literal reading of the sentence that each 89×89 minor belongs to one of those modules.

7 What this does and does not prove

7.1 What is proved

1. The 89×89 determinant specified in `results/certificates/lemma19_89_minor_weight_audit.json` is nonzero in characteristic zero.
2. Its torus weight is $(60, 59, 59, 140, 123, 126, 119, 115)$.
3. This weight cannot occur in any of the 15 certified predecessor irreducibles.
4. Therefore the printed universal statement about all 89×89 minors is false if read literally.
5. The only currently certified degree-89 Schur-highest weights that can feed $S_\nu W$ through $S_\lambda W \otimes S_\mu W$ are the 15 listed above; this is recorded in `lemma19_minimality_relevant_modules.json`.

7.2 What is not proved

1. This does not prove that Lee's claimed degree-90 generator is nonminimal.
2. This does not prove that $S_\nu W$ is absent from the span of all 90×90 maximal minors.
3. This does not prove that the theorem of the paper is false.
4. This does not rule out a corrected interpretation such as: "each relevant 89×89 component that can contribute to $S_\nu W$ lies in one of these 15 modules."
5. This does not construct the actual embedded $S_\lambda W$-copies inside $\mathrm{Sym}^{89}(S_\mu W)$.
6. This does not perform Lee's unpublished test that none of those embedded copies lies in J_8.

The last point is important. The displayed 89×89 minor has dominant first part 140, so it is not even contained in ν, whose first part is 133. Hence this particular 89×89 module is not an obvious source for the target $S_\nu W$ after multiplication by the degree-one coordinate representation. It refutes the paper's broad wording, but it does not by itself exhibit a lower-degree generator that produces the target degree-90 copy.

8 How it could still affect minimality

Lee's intended minimality argument must show that the target degree-90 copy of $S_\nu W$ is not generated by degree-89 elements of J_8. A complete certificate should identify the multiplication map

$$R_1 \otimes (J_8)_{89} \longrightarrow (J_8)_{90}$$

and prove that the relevant $S_\nu W$-projection is not in the image.

The current 89×89 counterexample matters because it shows that the published bookkeeping of degree-89 minors is not literally complete. This creates two possible outcomes:

- **Repair path.** The author only needed the 15 modules that can map to $S_\nu W$, and the extra 89×89 minors are irrelevant to the target projection. Then the paper's wording is wrong, but the construction may still be repairable.

- **Failure path.** The missing degree-89 bookkeeping hides a nonzero image into the claimed degree-90 constituent, making the purported degree-90 generator nonminimal. Proving this requires an explicit projection or multiplication certificate, not just the weight obstruction in this report.

9 Evidence table

Item	Source of evidence	Status
Matrix shape 90×115 and 1410 sparse entries	SageMath generator plus Rust, Singular, Oscar, Magma verifiers	**Computation/certificate**
Full-rank 90×90 specialization	`lemma19_nonzero_minor.json`; determinant 970351 modulo 1000003	**Computation/certificate**
Nonzero 89×89 minor	`lemma19_89_minor_weight_audit.json`; determinant 421057 modulo 1000003	**Computation/certificate**
Weight formula for coordinates and determinants	Torus action on $p_{r,st}$; additive row/column factorization	**Theorem plus checked bookkeeping**
List of 15 predecessor partitions	Sage LR enumeration plus independent Rust LR-tableau verifier	**Computation/certificate**
Minimality-relevant $V = S_\lambda W$ candidates	`lemma19_minimality_relevant_modules.json`; same 15 λ's, with $c^{\nu}_{\lambda,\mu} = 1$ and tensor-chain pointers	**Computation/certificate, tensor level only**
Weight exclusion from the 15 modules	Highest-weight theory: weights of $S_\lambda W$ are dominated by λ	**Theorem**
Copies inside $\mathrm{Sym}^{89}(S_\mu W)$	Requires plethysm/symmetrization multiplicity-space data	**Open**
Degree-90 minimality	Requires the $S_\nu W$-projection of $R_1(J_8)_{89} \to (J_8)_{90}$	**Open**
Lee's theorem and Haiman counterexample	Requires Lemma 19 plus downstream regularity and geometry audit	**Open**

10 Reproduction commands

The relevant local commands are:

```
./scripts/validate-all.sh

python3 artifacts/common/analyze_minor_weight.py

cargo run --quiet --locked \
  --manifest-path artifacts/rust/Cargo.toml
```

The important certificate files are:

- `results/certificates/lemma19_89_minor_weight_audit.json`;
- `results/certificates/lemma19_claimed_weight_raising_audit.json`;
- `results/certificates/lemma19_predecessor_partitions.json`;
- `results/certificates/lemma19_minimality_relevant_modules.json`;
- `results/certificates/lemma19_nonzero_minor.json`.

11 Recommended next checks

1. Replace the phrase "all 89×89 minors" by the precise representation-theoretic subspace that can contribute to $S_\nu W$, if such a repair exists.
2. Compute the $S_\nu W$-isotypic projection of the maximal-minor span.
3. Compute the $S_\nu W$-projection of the multiplication image from degree 89.
4. Decide whether the target degree-90 copy is outside that image (minimality) or inside that image (nonminimality).
5. Only after that, return to the geometric and regularity steps of the paper.

12 Current verdict

The current verdict remains **open audit**. The second literal assertion in Lemma 19 is refuted as written, and this is a natural place to investigate the possible nonminimality of the claimed degree-90 generator. However, the available evidence does not yet prove nonminimality. The next decisive object is not another isolated torus weight, but the actual $S_\nu W$-projection of the degree-90 minors and the multiplication image from degree 89.